Problem Set 2

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Problem 1: *The centralizer of an element.*
Let G be a group and let and let $a \in G$. The centralizer of a is the set of all elements that commute with a:

$$C(a) := \{ g \in G : ga = ag \}$$

(a) In the group S_5, let $\sigma = (135)$. List all the elements of the centralizer $C(\sigma)$.

Solution:

To find all elements in the centralizer of σ, we must find all elements, $\tau \in S_5$ such that $\sigma\tau = \tau\sigma$.

Helping us in that endeavour is Proposition 2.6., which states that if τ satisfies $\text{supp}(\sigma) \cap \text{supp}(\tau) = \emptyset$, then τ and σ commute. Clearly, σ permutes 1, 3, and 5 while leaving 2 and 4 fixed. So, by definition of the support, $\text{supp}(\sigma) = \{1, 3, 5\}$. Therefore, any τ satisfying $\text{supp}(\tau) \subseteq \{2, 4\}$ must commute with σ. Evidently, there are only two such τ and so we can conclude that at least:

$$\tau = (2, 4),\ \tau = ()$$

must permute with σ and so belong to the centralizer $C(\sigma)$.

(b) Prove that $C(a) \leq G$ for all $a \in G$.

Solution:

First, let $a \in G$ be arbitrary. To prove that $C(a) \leq G$, we must show that $C(a)$ is a subgroup of G, which we can do by the one-line subgroup test!

The one line subgroup test states that if $xy^{-1} \in C(a)$ for all $x, y \in C(a)$, then $C(a)$ must be a subgroup of G. So, to that end, letting $x, y \in G$ be arbitrary, I will now proceed to show that $(xy^{-1})a = a(xy^{-1})$, hence meaning that $xy^{-1} \in C(a)$ and concluding the proof.

By definition of $C(a)$, we know that $x, y \in a$ implies that $xa = ax$ and $ya = ay$. We also know, since G is a group and $C(a)$ is clearly a subset of G, that y must have an inverse in G. Letting $y^{-1} \in G$ be y's inverse, we can see that:

$$\begin{aligned}
y^{-1}a &= y^{-1}(a(1)) && \text{(By definition of the identity)} \\
&= y^{-1}(a(yy^{-1}) && \text{(By definition of inverses)} \\
&= y^{-1}((ay)y^{-1}) && \text{(By associativity)} \\
&= (y^{-1}(ay))y^{-1} && \text{(By associativity)} \\
&= (y^{-1}(ya))y^{-1} && \text{(Since } y \in C(a)\text{)} \\
&= ((y^{-1}y)a)y^{-1} && \text{(By associativity)} \\
&= ((1)a)y^{-1}y^{-1} && \text{(By definition of inverses)} \\
&= ay^{-1} && \text{(By definition of the identity)}
\end{aligned}$$

Therefore, $y^{-1}a = ay^{-1}$, implying $y^{-1} \in C(a)$ and thereby making $xy^{-1} \in C(a)$ a corollary because:

$$\begin{aligned}
(xy^{-1})a &= x(y^{-1}a) && \text{(By associativity)}\\
&= x(ay^{-1}) && \text{(Since } y \in C(a)\text{)}\\
&= (xa)y^{-1} && \text{(By associativity)}\\
&= (ax)y^{-1} && \text{(Since } x \in C(a)\text{)}\\
&= a(xy^{-1}) && \text{(By associativity)}
\end{aligned}$$

Immediately follows. Thus, by the one-line subgroup test, $C(a)$ is indeed a subgroup of G.

(c) Two friends, Goofus and Gallant, are arguing about the following claim:

"If $a, b \in G$, then $C(a) \cap C(b) = C(ab)$"

Goofus thinks "$\subseteq$" is true, but Gallant thinks "$\supseteq$" is true. Who is correct? Prove one inclusion, and give an explicit counterexample showing that the other inclusion is false in general.

Solution:

I contend, and will proceed to show, that $\subseteq$ is indeed the only correct inclusion. To do this, I intend to formally prove that $(C(a) \cap C(b)) \subseteq C(ab)$ for any arbitrary $a \in G$, for any arbitrary group G, and provide a specific group G with elements $a, b \in G$ such that $(C(a) \cap C(b)) \supseteq C(ab)$ doesn't hold, making Gallant's inclusion false in general.

(I) Prove that $(C(a) \cap C(b)) \subseteq C(ab)$.

To prove this inclusion, I will show that $x \in (C(a) \cap C(b))$ implies that $x \in C(ab)$. So, let any such x be arbitrary.

By definition